

Presentation 1:

- Links to presentation(s) and code(s) on GitHub:
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Resource%20allocation%20analysis/PhoneLineVariabilityAndLanguageAllCalls_Oct6_Sana.ipynb
 - First Link is where main code for language distribution and phone line data is. This was after figuring out two phone lines that led to English/Spanish main menus.
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Resource%20allocation%20analysis/PhoneLineVariability_Oct6_Sana.ipynb
 - The links above are to my code for resource allocation. There's some code and analysis that isn't mentioned in the presentation because there was not much that I learned from it.
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Resource%20allocation%20analysis/Resource_Allocation_Presentation_1_Oct7_Sana.pdf
 - This is the link to the slides for presentation 1. My updates are on slides 12-17.
- What did you do?
 - I focused on presentation one to get a better understanding of the data queues and the routes callers make to get to their destination. My two main focuses were on phone line distribution and language choices. I was hoping to see if there were some large discrepancies in which lines were called more than others, especially during different times of the year/day/week. I was also hoping to see language distributions, however, I am still in the process of understanding the phone logs to identify which callers are selecting Spanish as their language or waiting for an interpreter at the language menu.
 - After Friday Mentor Meeting, was able to look at duration of calls between Spanish and English calls by filtering by inbound calls and grouping by correlation ID and finding averages of calls (Spanish and English) Spanish calls were by average 78 seconds longer. Why?
- How does it help the project?
 - This should help the project because it informs us and Legal Aid of how the current system is being used by customers and if their intended audience/clients are being served. (Spanish and other language-speakers). It can also help improve the menus by possibly finding redundancies (such as the choice of where to place the language menu)
- Issues faced (if any):
 - My main issue with looking at the language chosen for each call was finding the 'activity' or prompt where customers selected the language. According to the verbiage spreadsheet, there are WCC prompts in both Spanish and English. However I'm never seeing a single instance of any prompt that is in Spanish in both the CAR and All Call data. The only example is when customers select the staff directory option from the main menu in Spanish. At that point they've already selected Spanish from the first language selection menu, so why is the main menu not being shown as main_menu_spanish?
 - I'm also interested in understanding which phone numbers are being used. I'm referencing this spreadsheet to understand which phone number is to what department/organization, however I'm not seeing any of these numbers in the data? I'd like to better understand how to track this.

- Attempts to resolve issues (if any)
 - I attempted to resolve this issue by searching for different examples of Spanish being mentioned or any indication of what the caller selected in the language menu but I'm still confused about this.
 - Update: After Friday Mentor Meeting, was able to find some indication of Spanish/English calls in the All Cars Dataset. However, still would like some indication of more examples of Spanish queues (did people switch over from Spanish to English or vice versa later on in their call?)
- Issues resolved (if any)
 - N/A
- References (Mention if you built up on someone else's work)
 - N/A
- Next Steps:
 - I'm interested in looking at other ways of identifying English/Spanish calls, especially to look at different switches during a singular call. Maybe a Spanish speaker who requests Spanish at a later time of the call? Or similar instances. I also am interested in looking at other ways to compare the English and Spanish calls outside of the duration of calls.